

BHUPENDER SINGH

Technical Manager | Senior Software Engineer | AdTech & Distributed Systems

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PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer and Technical Manager with 6.5+ years of end-to-end experience building scalable distributed systems, monetization infrastructure, and high-traffic AdTech platforms. Deep expertise in header bidding ecosystems (Prebid.js), server-side Conversion APIs (Meta, Google, TikTok, Taboola), microservices architecture, and full-stack development with Next.js, React, TypeScript, Node.js, and Python. Proven track record of leading engineering teams, driving 15–20% revenue yield improvements, and architecting systems handling 800K–1M+ daily requests. Experienced in AI-enabled platforms using Gemini (Vertex AI), cloud infrastructure (AWS, GCP), and DevOps best practices (Docker, Kubernetes, CI/CD).

TECHNICAL SKILLS

Languages & Frontend: JavaScript (ES2023), TypeScript, Python, HTML5, CSS3, React 19, Next.js (SSR/ISR/App Router), Vue.js

Backend & APIs: Node.js, Fastify, Express, RESTful APIs, GraphQL, gRPC, WebSocket, Webhooks, OAuth 2.0, Token Rotation, Nginx, Apache

Databases & Messaging: MongoDB, Redis, SQL (MySQL/PostgreSQL), Kafka, Async Worker Queues, Data Analytics Dashboards

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB, CloudWatch, Cognito, IAM, VPC), GCP, Docker, Kubernetes, CI/CD Pipelines, Cloudflare CDN, Grafana, Uptime Monitoring

AI & LLM: Gemini (Vertex AI), Prompt Engineering, LLM Integration, AI-based Keyword & Content Generation

AdTech & Monetization: Google Ad Manager (DFP/GAM), Prebid.js, Header Bidding, AFS/AFG AdSense, Server-Side CAPI (Google, Meta, TikTok, Taboola, Outbrain), Google Ads API, Analytics API, AdSense Management API, SSP Integrations (OpenX, PubMatic, Magnite, OneTag, 33Across, MGID)

Architecture & Practices: Microservices, Distributed Systems, System Design, OWASP Security (XSS/CSRF, HTTPS/TLS, Token Rotation), GDPR-Aligned Tracking, A/B Testing, Code Review, Webpack, Babel, Jest

UI Libraries & State: Redux, Zustand, Material UI, Ant Design, Tailwind CSS, Bootstrap

PROFESSIONAL EXPERIENCE

Unibots | New Delhi, India | Oct 2019 – Present | 6.5+ Years

Full-service publisher monetization platform serving 100M+ monthly page views across global AdTech, analytics, and programmatic advertising products.

4 Technical Manager

Apr 2025 – Present

- **Led AFS & AFG AdSense integrations** across search and gaming platforms, optimizing Video Player and Prebid.js stack to improve ad yield by ~15–20% and reduce response latency by ~10–15%.
- **Architected scalable AdTech microservices** (Fastify, MongoDB, Redis, async worker queues) handling ~800K–1M RPS with rate limiting and structured error handling, maintaining ~99.9% uptime.
- **Designed and deployed** a Google Ads RAC-compliant media lifecycle microservice and unified Campaign Management System enabling multi-platform campaign creation, budget control, and real-time performance analytics.

- **Implemented server-side and web-based Conversion APIs** (Meta, Google Ads, Taboola, TikTok, Outbrain), improving attribution accuracy while ensuring privacy-compliant GDPR-aligned tracking.
- **Reduced API response time by ~10–15%** through MongoDB index optimization and Redis caching; improved page load performance by ~15% via CDN integration and Next.js SSR optimization.
- **Established CI/CD pipelines** with Docker and Kubernetes; implemented OAuth 2.0 authentication and OWASP-aligned API security (XSS/CSRF protection, token rotation), reducing infrastructure costs by ~15–20%.
- **Built AI-enabled analytics platforms** using Gemini (Vertex AI) for keyword and content generation, improving scalability and revenue outcomes for global publishers.

3 Tech Team Lead

Apr 2022 – Mar 2025

- **Led a cross-functional team of 6+ engineers** delivering AdTech and analytics products across high-traffic publisher platforms generating 100M+ monthly page views.
- **Architected header bidding infrastructure** using Prebid.js with custom timeout strategies and bid adapters, increasing publisher yield by 15–20% and reducing discrepancies by significant margins.
- **Designed scalable microservices** architecture handling ~1K RPS with Redis caching, async job processing, and horizontal scaling patterns.
- **Built Google Ad Manager (GAM) analytics dashboard** used by internal teams to optimize revenue performance across multiple geographies and verticals.
- **Standardized ad integrations** via custom NPM modules and shared component libraries, reducing new publisher onboarding time by ~30%.
- **Integrated multiple SSP bidding APIs**: OpenX, PubMatic, Magnite, OneTag, 33Across, and MGID — improving partner coverage and fill rates across all publisher inventory.
- **Mentored junior and mid-level developers**, enforced structured code review processes, and introduced testing standards that improved release stability and reduced production incidents.

2 Sr. Web Developer

Oct 2020 – Mar 2022

- **Developed high-performance monetization widgets** deployed across multiple publisher sites, significantly improving ad viewability scores and user engagement metrics.
- **Implemented Google Ad Manager (GPT) and Prebid.js integrations** end-to-end, reducing average ad latency by ~20% through asynchronous loading and lazy rendering strategies.
- **Diagnosed and resolved large-scale ad-serving discrepancies** between SSP reports and GAM, improving reporting accuracy and strengthening partner trust.
- **Optimized frontend performance** using code splitting, lazy loading, and efficient React rendering patterns, resulting in measurable improvements in Core Web Vitals and page load times.

1 Web Developer

Oct 2019 – Sep 2020

- **Built interactive quiz and gaming modules** integrated with Facebook authentication APIs, increasing average user engagement duration by measurable margins.
- **Developed AMP-compatible ad widgets** compliant with Google AMP specifications, improving mobile page performance and monetization efficiency on low-bandwidth connections.
- **Automated dynamic content switching logic** for seasonal and event-based campaigns, enhancing user retention during off-peak traffic periods.

Front-End Intern | Tisaaw Services, Noida | Jun 2019 – Sep 2019

- Resolved UI bugs and delivered new feature modules for ShareCreators.com using Vue.js and Bootstrap.
- Developed login and authentication UI components with Vue.js, CSS3, and Bootstrap, contributing to improved onboarding UX.

KEY PROJECTS

Indulights – B2B Commerce Platform | *INDU LIGHTS PVT LTD (indulights.in)* | Aug – Dec 2025

- **Owned full product architecture** of a client-facing B2B e-commerce platform with RBAC-based admin control, custom role permissions, and multi-tenant support.
- **Built scalable inventory management** and custom order workflows with automated invoice generation, reducing manual processing effort significantly.
- **Integrated Meta Business API** for bulk messaging and real-time order notifications, improving customer communication throughput.
- **Optimized MongoDB queries and Redis caching**, reducing average API response latency by ~20% at peak load.
- **Improved dashboard performance by ~15%** through Next.js SSR/ISR optimizations, reducing time-to-interactive for data-heavy admin views.
- **Implemented rate limiting, secure JWT authentication, and OWASP-compliant security practices** across all API endpoints.

EDUCATION

Bachelor of Technology – Computer Science & Engineering

Bundelkhand Institute of Engineering & Technology, Jhansi, Uttar Pradesh | 2015 – 2019